RaceDay API - Implemented Endpoints (Part 2)

PROG6212 Programming 2B | ASP.NET Core Web API (.NET 8) | All routes start with /api/

How to read this document. Role Required: *None* = public; *Any* = any logged-in user; otherwise only that role. *(owner)* means the Organiser must also own the event concerned; another Organiser receives 403.
Authentication. After login the API keeps the user ID and role in a server-side session and a cookie; Swagger sends the cookie automatically. Anyone can register as a Participant; registering as an Organiser requires the organiser invite code.
Validation. Requests with invalid fields return 400 Bad Request. Protected endpoints return 401 when not logged in and 403 for the wrong role.

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/Auth/register`	Creates a new Organiser or Participant account. Passwords are hashed. Organisers must supply the invite code.	**None (public)**	email, password (min 8), firstName, lastName, role (Organiser | Participant). **Organiser:** organisationName, phoneNumber, inviteCode. **Participant:** dateOfBirth (past date).	**200 OK** - user summary **400** - missing/invalid details **403** - invite code missing or wrong **409** - email already registered
POST	`/api/Auth/login`	Logs in, signs the user in and starts a server-side session storing the user ID and role.	**None (public)**	email, password	**200 OK** - user summary, session cookie set **401** - wrong email or password
GET	`/api/Auth/me`	Returns the user ID and role held in the server-side session.	**Any (logged in)**	None	**200 OK** - { userID, role } **401** - not logged in / session expired
POST	`/api/Auth/logout`	Clears the server-side session and signs the user out.	**Any (logged in)**	None	**200 OK** - message

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/Profile`	Shows the logged-in user's own profile.	**Any (logged in)**	None	**200 OK** - profile (name, email, role, plus organisation/phone or date of birth/picture) **401** - not logged in **404** - user not found
PUT	`/api/Profile`	Updates the logged-in user's own profile.	**Any (logged in)**	firstName, lastName. **Organiser:** organisationName, phoneNumber. **Participant:** dateOfBirth (past date).	**200 OK** - message **400** - invalid details **401** - not logged in **404** - user not found

3. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/Events`	Lists all events, soonest first, with enrolment counts.	**None (public)**	None	**200 OK** - list of events
GET	`/api/Events/{id}`	Gets one event by ID.	**None (public)**	None	**200 OK** - event **404** - event not found
GET	`/api/Events/mine`	Lists the events created by the logged-in Organiser (dashboard).	**Organiser**	None	**200 OK** - list of events **401** - not logged in **403** - Participant
POST	`/api/Events`	Creates an event owned by the logged-in Organiser.	**Organiser**	name, description, eventDate, location, distance (0.1-1000 km), eventType (Run | Walk | Cycle)	**201 Created** - event **400** - invalid details **401** - not logged in **403** - Participant
PUT	`/api/Events/{id}`	Updates one of the Organiser's own events.	**Organiser (owner)**	name, description, eventDate, location, distance, eventType	**200 OK** - updated event **400** - invalid details **401** / **403** - not logged in, Participant or not the owner **404** - event not found
DELETE	`/api/Events/{id}`	Deletes one of the Organiser's own events.	**Organiser (owner)**	None	**200 OK** - message **401** / **403** - not logged in, Participant or not the owner **404** - event not found **409** - participants already enrolled

4. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	`/api/Categories`	Lists all categories for all events.	**None (public)**	None	**200 OK** - list of categories
GET	`/api/Categories/event/{eventId}`	Lists the categories of one event.	**None (public)**	None	**200 OK** - list of categories **404** - event not found
POST	`/api/Categories`	Adds an age or distance category (e.g. 10km, Under 20) to the Organiser's own event.	**Organiser (owner)**	eventID, name, description (optional)	**200 OK** - category **400** - invalid details **401** / **403** - not logged in, Participant or not the owner **404** - event not found

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	`/api/Categories/{id}`	Updates a category on the Organiser's own event. A category cannot be moved to another event.	**Organiser (owner)**	eventID (must stay the same), name, description (optional)	**200 OK** - category **400** - invalid details / event changed **401** / **403** - not logged in, Participant or not the owner **404** - category not found
DELETE	`/api/Categories/{id}`	Deletes a category from the Organiser's own event.	**Organiser (owner)**	None	**200 OK** - message **401** / **403** - not logged in, Participant or not the owner **404** - category not found **409** - participants enrolled in it

5. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/Enrolments`	Enters the logged-in Participant into an event in a chosen category. Status always starts as Pending.	**Participant**	eventID, categoryID	**200 OK** - enrolment (event, category, status) **400** - invalid event/category, or event already took place **401** / **403** - not logged in or Organiser **409** - already enrolled in this event
GET	`/api/Enrolments/my`	Lists the Participant's own enrolments with event, category and status.	**Participant**	None	**200 OK** - list of enrolments **401** / **403** - not logged in or Organiser **404** - participant profile not found
GET	`/api/Enrolments/event/{eventId}`	Lists every enrolment for one of the Organiser's own events.	**Organiser (owner)**	None	**200 OK** - list of enrolments **401** / **403** - not logged in, Participant or not the owner **404** - organiser profile not found
PUT	`/api/Enrolments/{id}/status`	Sets an enrolment's status for an event the Organiser owns.	**Organiser (owner)**	status (Pending | Confirmed | Cancelled)	**200 OK** - updated enrolment **400** - invalid status **401** / **403** - not logged in, Participant or not the owner **404** - enrolment not found

6. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/Results`	Records the finish time and finishing position for an enrolment.	**Organiser (owner)**	enrolmentID, finishTime (hh:mm:ss), finishingPosition (1 or more)	**200 OK** - result **400** - invalid details / cancelled enrolment **401** / **403** - not logged in, Participant or not the owner **404** - enrolment not found **409** - result already exists
GET	`/api/Results/my`	Lists the Participant's own results: event, date, category, finish time, position and total finishers.	**Participant**	None	**200 OK** - list of results **401** / **403** - not logged in or Organiser **404** - participant profile not found
GET	`/api/Results/event/{eventId}`	Lists all results for one of the Organiser's own events, best position first.	**Organiser (owner)**	None	**200 OK** - list of results **401** / **403** - not logged in, Participant or not the owner **404** - event not found
PUT	`/api/Results/{id}`	Corrects the finish time and position of a result.	**Organiser (owner)**	finishTime (hh:mm:ss), finishingPosition	**200 OK** - updated result **400** - invalid details **401** / **403** - not logged in, Participant or not the owner **404** - result not found
DELETE	`/api/Results/{id}`	Deletes a result.	**Organiser (owner)**	None	**200 OK** - message **401** / **403** - not logged in, Participant or not the owner **404** - result not found

Total: 4 authentication + 2 profile + 6 event + 5 category + 4 enrolment + 5 result = **26 endpoints**.